

scenarios. That is why testing is the most important stage of a software development process.

Testing is mandatory because we are human, and we all make mistakes. But some of these mistakes can lead to complete software failure. If you deliver the software without testing it first and it is error-prone, all the effort and money that went into the development of software will go in vain and your business may suffer big times.

Here are some of the key reasons that explain why testing is essential and should never be ignored:

- **Reliability:** Software testing is needed to find defects or errors that were made during the development stages to ensure the reliability of software.
- **Quality products:** Software testing must be done to ensure the delivery of quality products. Delivering good quality products in time help in gaining customer confidence and trust in the company.
- **Cost-effectiveness:** Early software testing saves time and money. Many stages are involved in a software testing process. If defects or errors are identified in the initial stages itself, then it costs much less to fix them. Else, it can be costly to fix them since you might have to plan again, redesign, reimplement, and retest the software.

What are the types of software testing?

There are several types of testing that you can do to ensure that the software is functioning as per the actual requirements. At a high level, there are two types of testing: manual and automatic. But, if you go deep down, Automation testing is further classified into several other types of testing such as unit testing, regression testing, system testing, integration testing, etc.

1. Manual Testing

Manual testing is done by manually interacting with the software without the use of any script or tools. In this testing, the tester assumes himself/herself as an end-user. He/She tests the software thoroughly to find any errors and unexpected behaviour in the software.

This type of testing is generally expensive and time-consuming as the tester needs first to set up the testing environment, write test-cases, and execute test cases himself. Manual testing can lead to errors as being human, testers can make mistakes, and it becomes difficult and complex to test each component of software manually. It also requires repeated human intervention.